

Familiar Faces

(prototype)

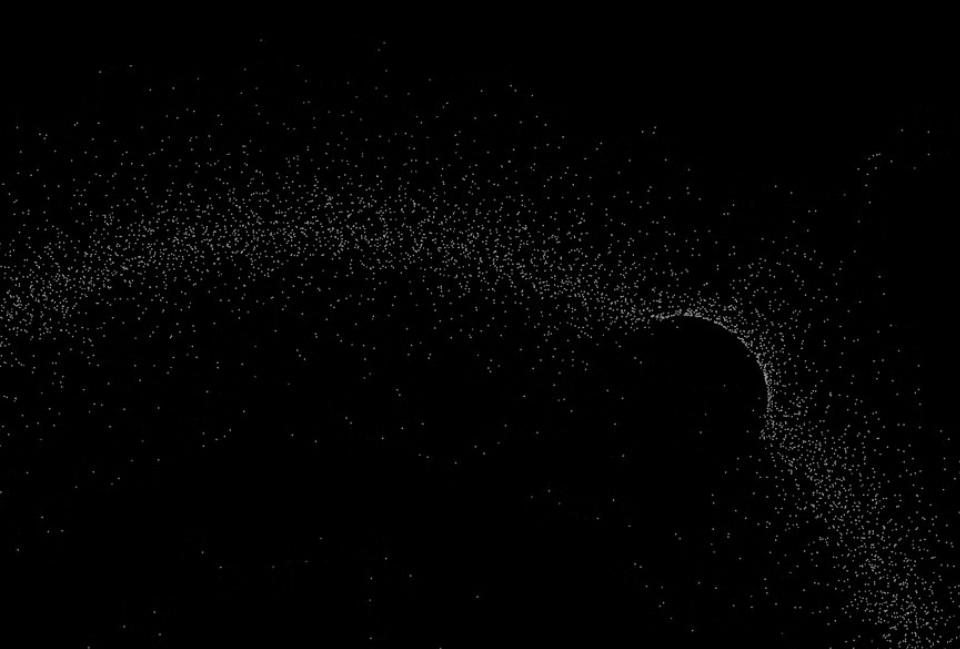


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The Problem

The Name Challenge

When meeting new people, remembering names can be a challenging task.

The Setting

This task can be especially challenging when meeting lots of new people in a short span of time, like in classrooms.



Image generated by Google Gemini: Prompt "An awkward conversation between two people where one guy can't remember the other's name"

The Problem



Social Impact & Psychology

Remembering names is more than courtesy—it avoids the "spotlight effect" where we over-perceive our own blunders.

Focus & Interaction

Failure in recall results in a loss of focus and significantly degrades the quality of personal interactions.

Trust & Team Building

Name recall is vital for building cohesive teams and is a proven method for increasing trust.

Environment Constraint

Traditional methods like "the name game" are ineffective in today's fast-paced environments.



Familiar Faces Offers a Technological
Solution to this Social Problem

The Solution: Face Detection

YOLOv26 Model

Face Detection is handled by a YOLOv26 nano object detection model trained on the Wider-Face dataset.

Performance Optimization

Detection runs every 10 frames to reduce computational cost, ensuring high performance on mobile devices.

Kalman Filter

A kalman filter runs constantly on the main UI thread to track faces in between detections at very little cost.

The Solution: Face Recognition

FaceNet Model

Facial Recognition is handled by the FaceNet model by Google through the Deeper Face library.

128D Embedding

Faces are embedded into 128 dimensional space by the FaceNet model.

KNN Matching

A KNN algorithm then checks to see if the new face matches an existing contact.

UI & Database

Results return to the UI thread; unknown embeddings are stored in the contact database.

The Solution: Face Storage

SQLite Database: Embeddings and contacts are stored in a SQLite database.

Contacts Table: Known contacts are stored with name, profile picture, number of times seen, and last seen date.

Embeddings Table: Specialized table for storing facial contact embeddings.

Unknown Contacts: Dedicated table for tracking unknown face data.

unknown_contacts	
id	integer
last_seen	timestamp
image	blob
embedding	blob NN

contacts	
id	integer
name	text NN
encounter_count	integer NN
image	blob NN
last_seen	timestamp

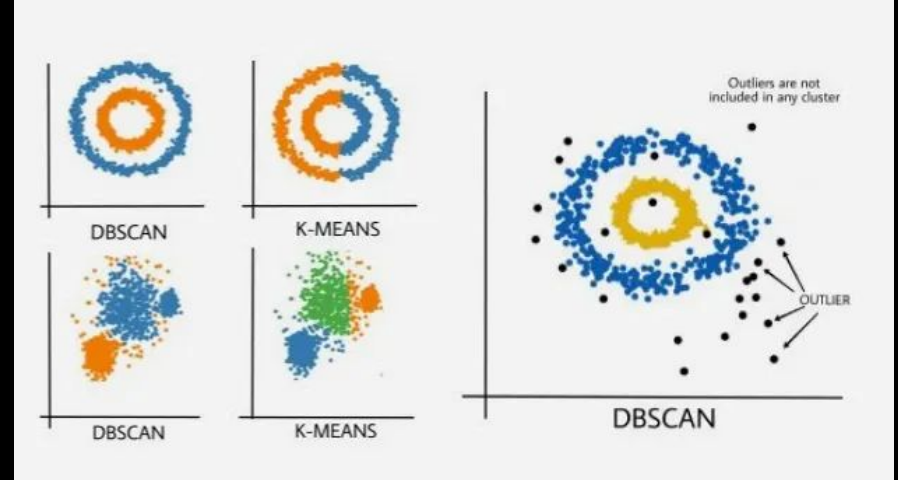
embeddings	
id	integer
contact_id	integer NN
created_at	timestamp
embedding	blob NN

The Solution: Face Storage

DBSCAN Clustering: Periodically, the algorithm is run on the "unknown contacts" table to identify clusters of new faces based on data frequency (density)..

Density-Based Logic: Points are grouped if the density reaches a defined threshold, classifying them as a single face cluster.

UI Integration: Once identified, clusters are passed to the main UI where users can name and store them.



Debomit Dey, "DBSCAN Clustering in ML - Density based clustering," Kmeans vs DBSCAN graphic, <https://www.geeksforgeeks.org/machine-learning/dbscan-clustering-in-ml-density-based-clustering/>, 19:59:04+00:00.

Project Demo

Please standby while I prepare the demo

Results: Quantitative

Model Performance (YOLOv26 nano):

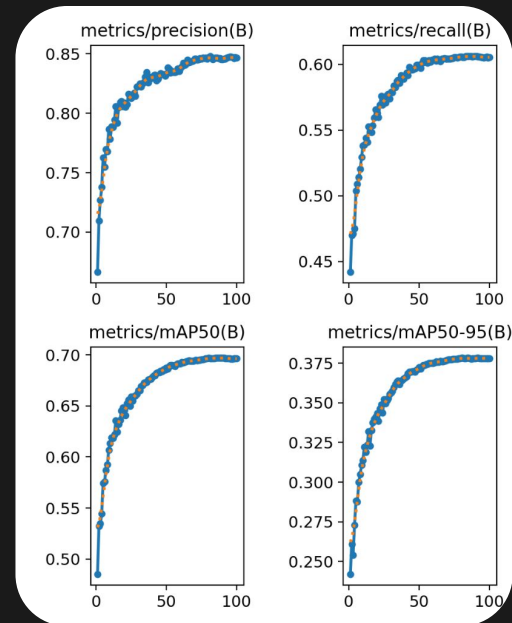
Achieved a **precision of 84%** and **recall of 60%**.

Mean Average Precision (mAP):

mAP@0.5-0.95 measured at **38%**. Given the complexity of the dataset, this represents very strong performance.

WIDER FACE Dataset:

Consists of **32k images** and **393,703 annotated faces**, making it one of the most challenging available.



Results: Qualitative

User Acceptance Testing

Tests were used to collect feedback on the real-world performance of Familiar Faces.

Visual Experience

The pop-up was noted as unobtrusive and disappeared into peripheral vision easily.

Detection Range

Detection range exceeded expectations, providing valuable feedback for future iterations.

What's Next for Familiar Faces

Mobile Framework Integration

Converting the prototype pipeline to mobile frameworks to enhance accessibility and performance.

Android XR Framework

Extending the technology to the Android XR framework for seamless use on wearable devices.

Detection Optimization

Continuous tuning and testing to decrease false negative detections and improve accuracy.



Thank you